

The information is gathered according to the following list of topics for each of Bacon and Popper:

- Context of the author
 - The key idea
 - The enemy/critique
 - The proposed method
 - The status of truth
 - The role of observation/experiment
 - The main criticisms
 - Others
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BACON

Context of the author

Bacon was of a time right after **Medieval scholasticism and Aristotelian logic**, schools that believed knowledge is to be generated from conceptually precise logical analysis. This was a mostly intellectual way of seeing knowledge generation. Also the foundations to begin logical analysis usually came from **authority**, which at the time usually meant the church but could also mean Aristotle or other known writers such as Thomas Aquinas.

The Key Idea

Knowledge cannot be generated through purely intellectual means, it must be based in controlled observation, experimentation and **induction**.

The enemy/critique

Scholasticism and Aristotelian syllogistic logic. Bacon's main work is called *Novum Organum* (new instrument) in contrast to Aristotle's *Organum* (instrument) which reinforces that Bacon say his ideas as superseeding Aristotle's.

Bacon thought that **deductive methods can only reveal knowledge that was contained in the assumed premise, but cannot by itself generate new knowledge**

The proposed method

Instead of using deduction and going

general $\rightarrow$ particular

Bacon proposed **Induction** which is a method for going particular for going

particular $\rightarrow$ general

This is the main contribution of Francis Bacon to the philosophy of science. Induction is, at its core, the following reasoning: **Let S be a set of objects, then if I see that sufficiently many objects $x \in S$ satisfy a property P we can say that all objects of S satisfy P .** As an example induction would say that repeated observation that metals expand when heated imply that all metals expand when heated.

There are clear problems with the above logic but we shall treat them latter. I want to emphasize that there are subtleties to induction not conveyed in the above description. For instance, no amount of repeated observations that **copper** expands when heated can imply that **all metals** expand when heated. For the inductive conclusion to be valid the observations must be done covering a wide variety of scenarios and under a controlled environment. Bacon did not preach mindless observation.

The status of Truth

Truth is gradually discovered using controlled observation, experiment and induction. **Bacon is fairly optimistic about the certainty of the knowledge obtained by his methods.** He believed good methodology can correct human weakness.

Role of observation/experiment

Absolutely central. Observation and experiment is the foundation of knowledge to Bacon.

The main criticisms

Induction has several flaws. For starters, what does it mean to observe *sufficiently many x satisfy P* ? For sure this depends on the cardinality of S but how so? For instance, if S has one element, one observation surely suffices by what if S is a very very large set or, worst, a set of unknown size?

The truth is that until every element of S is verified manually to satisfy P we cannot say with certainty that all $x \in S$ satisfy P . At best we can give a probabilistic assessment of the likelihood of all $x \in S$ satisfying P *if the size of S is known*, which makes no knowledge attained via induction certain especially when the size of S isn't known (which occurs more often than not). However I should also mention that the probabilistic assessment given above was **NOT** done by Bacon and only came many years latter.

In general the jump from finitely many observations to the general conclusion is problematic and became a major point of attack by Karl Popper and David Hume.

Others

I will use this section to talk about the four **Idols**. Bacon did not believe humans are naturally rational observers, he believed we are subject to many biases and *idols*. Bacon's categorized the idols into four distinct categories:

1. **Idols of the Tribe:** Errors common to humanity as a species. Humans seek patterns too eagerly and prefer confirmation over contradiction.
 2. **Idols of the Cave:** Errors of the individual psychology (their mental cave). Two individuals with contradicting hypothesis can observe the same phenomenon and conclude the phenomenon supports their hypothesis.
 3. **Idols of the Marketplace:** Errors generated by imprecision of language and thought. The use of vague undefined terms leads to confusion.
 4. **Idols of the Theatre:** Blind acceptance of intellectual systems. Mixing the theatre for reality itself.
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POPPER

Context of the author

The main topics here are

- The success of modern physics (mainly Einstein's relativity)
- The crisis of induction
- Psychoanalysis
- Logical positivism
- Marxism

Popper was very impressed by Einstein's theory of relativity because it was bold and predicted falsifiable claims, to the contrast of other know theories at the time, such as psychoanalysis and Marxism. This made the *demarcation problem*, delimiting what is pseudoscience and what isn't, an important part of Popper's work.

The key idea (s)

Science advances through bold conjectures and refutations, not through verification.

A theory is scientific if, and only if, it is falsifiable.

Importantly the falsifiability must exist only in theory but possibly not in practice at the current moment. For instance, the claim that in the end of 2026 the sky will turn green is not verifiable as of May 2026 but will eventually be verifiable so is an affirmation inside the realm of science.

The enemy/critique

Induction: Bacon accepts Hume's critique of induction and believes it is logically unsalvageable.

Verificationism: Bacon did not believe universal theories could be logically justified and claims like *every swan is white* cannot be verified. But such a claim is falsifiable so it can be verified to be false, which generates an asymmetry that is foundational to his work

Pseudoscience: Theories that are unfalsifiable but claim to generate scientific knowledge. Popper had in mind psychoanalysis and Marxism.

Proposed method

Instead of verification, Popper proposes a method of

problem → hypothesis → severe testing → refutation or survival

Critically, Popper never claimed a theory was *verifiably true*, only that it is *not verifiably false yet*. Which leads directly to Popper's view of truth and knowledge.

The status of truth

Compared to Bacon, Popper is radically pessimistic when it comes to the status of truth. **He believes no scientific theory can be proven true, they can only be proven false.** However, the more severe the testing the theory has gone through and survived the more credible the theory is.

Role of observation/experiment

Popper sees observation as crucial just like Bacon but in a different way. **Popper considered observation and experimentation to be the main way to prove a theory false** and since falsifiability was central to Popper's theory of science so was observation and experimentation.

The main criticisms

Duhem-Quine problem: *Unambiguous falsification is impossible.* Say the current scope of human knowledge predicted an event but it did not happen. Knowing which theory should be falsified by this incorrect prediction is essentially impossible. For instance if Newtonian mechanics predicted a planetary orbit incorrectly it may not be the fault of the theory itself but of

the measurement equipment or, worse, of the assumptions used in the prediction. For instance there is an unaccounted for planet that altered the planetary orbit being predicted.